

Jihyun Lee, Ph.D.

CURRICULUM VITAE

Research Professor
Laboratory of Hearing, Balance and Integrated Neuroscience
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Education and Training

05/2015 – 10/2017	Mayo Clinic, Department of Neurologic Surgery	Postdoctoral
03/2014 – 04/2015	Institute for Basic Sciences (KAIST)	Postdoctoral
03/2006 – 02/2014	Ajou University, Neuroscience	Ph.D.
03/2004 – 02/2006	Seoul National University, Pharmacy	M.S.
03/2000 – 02/2004	Pusan National University, Microbiology	B.S.

Research Experience

06/2018 – Present	Research Professor Hallym University College of Medicine, Anyang, Republic of Korea Laboratory of Hearing, Balance and Integrated Neuroscience Ear and Interaction Center, Doheun Institute for Digital Innovation in Medicine (D.I.D.I.M.) Laboratory of Dr. Hyo-Jeong Lee <u>Project1</u> : the Digit in Noise task for people with mild cognitive impairment using EEG system <u>Project2</u> : the sound localization task for single sided deafness using EEG system <u>Project3</u> : The perception of musical emotion in people with hearing loss <u>Project4</u> : Audio-visual cross modality <u>Lab Management</u> : Directed the installation of custom soundproof booth panels and integrated EEG testing environments
05/2015 – 10/2017	Postdoctoral Research Fellow Mayo Clinic, Rochester, MN, USA Laboratory of Dr. Su-Youne Chang

Project1: Harmaline-induced tremor with anesthetized swines
Project2: The effect of deep brain stimulation of ventrolateral thalamus in harmaline-induced tremor rodent models

03/2014 – 04/2015

Postdoctoral Research

Institute for Basic Sciences in KAIST, Daejeon, Republic of Korea
Laboratory of Dr. Min Whan Jung

Project: Neural activity of interval timing in posterior thalamus and CA1 hippocampus using single unit recording

03/2006 – 02/2014

Ph.D. Research

Ajou University School of Medicine, Suwon, Republic of Korea
Laboratory of Dr. Min Whan Jung

Project: Temporal information is conveyed by multiple neural processes in parallel using single unit recording

03/2004 – 02/2006

M.S. in Pharmacy

Seoul National University of Manufacturing Pharmacy,
Seoul, Republic of Korea
Laboratory of Dr. Byung Hee Han

Project: Anti-inflammatory Effects of Apigenin on Lipopolysaccharide-induced Activation of Microglia is mediated by Inhibition of the NF-KB pathway in BV-2 cells

Technical Skills / Expertise

Data Analysis & Programming: MATLAB

Neurophysiological Techniques: EEG data processing, Single-Unit Recording

Experimental Design: Audiovisual integration, Musical emotion perception paradigms, Sound Localization paradigms, Digit-in-Noise Task

Funding

06/2023 – 05/2026

Basic Science Research Program through the National Research Foundation of Korea (NRF), funded by the Ministry of Education

Project Title: Audio-visual cross-modal reorganization on the perception of musical emotions in individuals with hearing loss

Role: PI

Total Funding: KRW 210,000,000

09/2020 – 08/2023

Basic Science Research Program through the National Research Foundation of Korea (NRF), funded by the Ministry of Education

Project Title: Investigating neurophysiological biomarkers of aberrant musical perceived emotions affected by frequency-specific hearing loss

Role: PI

Total Funding: KRW 135,000,000

Honors / Awards

- 2022** Travel Award: International Conference on Auditory Cortex
- 2020** Poster Award: The 7th Korean Society for EEG and Neurophysiology
- 2010** Poster Award: The 13th Annual Meeting of the Korean Society for Brain and Neural Science

Peer-reviewed Publications

- Lee J**, Han JH, Lee H-J (2026) Cortical responses to the lack of high-frequency cues in musical emotion perception *Scientific Reports*
- Jeon S-I, **Lee J**, Han JH, Jeong D, Lee H-J (2026) An Automated System for Sound Localization testing in Hearing-Impaired Listeners *JoVE*
- Lee J**, Han JH, Lee H-J (2025) Musical Training and Perceptual History Shape Alpha Dynamics in audiovisual Speech Integration *Brain Sciences*
- Han JH, Jo S, Jeon S-I, **Lee J**, Lee JH, Lee H-J (2025) Evaluation of the Korean digits-in-noise test to classify the degree of hearing loss *Int J of Audiol*
- Lee J**, Han JH, Lee H-J (2023) Development of novel musical stimuli for investigating the perception of musical emotions in individuals with hearing loss *J Korean Med Sci*
- Shim L*, **Lee J***, Han JH, Jeon H, Hong S-K, Lee H-J (2023) Feasibility of Virtual Reality-Based Auditory Localization Training with Binaurally Recorded Auditory Stimuli for Patients with Single-Sided Deafness *Clin Exp Otorhinol* (* co-first author)
- Han J-H, **Lee J**, Lee H-J (2023) The effect of noise on the cortical activity patterns of speech processing in adults with single-sided deafness. *Front Neurol*
- Han J-H, **Lee J**, Lee H-J (2023) Attentional modulation of auditory cortical activity in individuals with single-sided deafness. *Neuropsychologia*
- Han JH, **Lee J**, Lee HJ (2021) Ear-specific hemispheric asymmetry in unilateral deafness revealed by auditory cortical activity *Front Neurosci*
- Lee J**, Han JH, Lee HJ (2020) Long-term musical training alters auditory cortical activity to the frequency change *Front Hum Neurosci*
- Han JH, **Lee J**, Lee HJ (2020) Noise-induced change of cortical temporal processing in Cochlear Implant Users *Clin Exp Otorhinolaryngol*
- Han JH, Yi DW, **Lee J**, Chang WD, Lee HJ (2020) Development of a smartphone-based Digits-in-Noise test in Korean: a Hearing screening tool for speech perception in noise *J Korean Med Sci*

Lee J, Chang SY (2019) Altered primary motor cortex neuronal activity in a rat model of harmaline-induced tremor during thalamic deep brain stimulation *Front Cell Neurosci*

Lee J, Jo HJ, Kim I, **Lee J**, Min HK, In MH, Knight E, Chang SY (2019) Mapping BOLD activation by pharmacologically evoked tremor in swine *Front Neurosci*

Lee J, Han JH, Lee HJ (2019) Auditory cortical temporal processing and hemispheric asymmetry revealed by N1 Dipole Source Activity in adult cochlear implant users *Korean J Otorhinolaryngol-Head Neck Surg*

Lee J, Han JH (2019) The effect of Noise on Cortical Auditory Evoked Potential in Cochlear Implant Users *Audiol Speech Res.*

Lee J, Kim IY, Lee J, Knight E, Cheng L, Kang SI, Jang DP, Chang SY. (2018) Development of Harmaline-induced tremor in a Swine Model *Tremor Other Hyperkinet Mov.*

Han J, **Lee JH**, Kim MJ, Jung MW. (2013) Neural activity in mediodorsal nucleus of thalamus in rats performing a working memory task. *Front Neural Circuits* (co-first authorship)

Kim J, Ghim JW, **Lee JH**, Jung MW. (2013) Neural correlates of interval timing in rodent prefrontal cortex. *J Neurosci*

Manuscripts in Progress

Kim SY[†], **Lee J**[†], Lee YE, Lee H-J*, and Lee M* Frequency-Aware Emotional EEG Decoding under Auditory Spectral Degradation (Under Review, *Submitted to Affect Comp*; [†] co-first author)

Jo S*, **Lee J***, Lee H-J Differential effects of aging and high-frequency hearing loss on the perception of meter, harmony, and timbre in music (In preparation; * co-first author)

Han JH*, **Lee J***, Lee H-J Brain oscillations in hearing loss and mild cognitive impairment during speech-in-noise tests (In preparation; * co-first author)

Patent

Han J-H, Lee H-J, **Lee J**, Lee D, (2025) Registration (제 10-2778770 호) 소리 방향성 분별능의 자동 검사 방법 및 시스템 (Automated Method and System for Testing Sound Localization Ability)

Presentations

Jo S*, **Lee J***, Lee H-J (2026) Aging and high-frequency hearing loss differently affect music perception across meter, harmony, and timbre, Politzer, Poster (accepted)

Lee J, Han J-H, Lee H-J (2026) How musical training and perceptual history modulate alpha-band dynamics underlying audiovisual speech integration, World Congress of Audiology, Poster

Lee J (2025) Seeing Ears, Hearing Eyes, and the Brain, Department of Biomedical Software, Catholic University, Invited talk

Lee J, Han J-H, Jo S, Lee H-J (2025) Musical training and Prior Experience in Multisensory Speech Perception: Neutral Insights from the McGurk Effect, International Conference on Auditory Cortex, Poster

Lee J (2025) Musical expertise modulates auditory cortical activity and audiovisual speech integration, Korean Society of Otorhinolaryngology-Head and Neck Surgery, Oral

Lee J, Han J-H, Lee H-J (2024) Frequency-dependent audibility modulates alpha activity responses to perceived emotions in music, Society for Neuroscience, Poster

Lee J, Han J-H, Lee H-J (2024) High-frequency audibility affects alpha activity response to the perception of musical emotion, Organization for Human Brain Mapping, Poster

Lee J (2024) Music and Emotion. Daegu Catholic University, Department of audiology and speech-language pathology, Invited talk

Lee J (2024) Alpha Activity response to the perceived emotions is affected by frequency-dependent audibility. Korean Society of Otorhinolaryngology-Head and Neck Surgery, Oral

Lee J, Han J-H, Lee H-J (2023) Cortical activity response to perceived emotions of music dependent on loss of high- or low- frequency audibility, International Conference on Music Perception and Cognition, Poster

Lee J, Han JH, Lee H-J (2022) Long-term musical training affects auditory cortical activity to audiovisual speech perception. International Conference on Auditory Cortex, Poster

Lee J (2022) Music and Emotion. Daegu Catholic University, Department of audiology and speech-language pathology, Invited talk

Han J-H, **Lee J**, Lee H-J (2022) Cortical encoding of speech in noise in single-sided deafness. Korean Society of Otorhinolaryngology-Head and Neck Surgery, Poster

Han J-H, **Lee J**, Park E, Lee H-J (2022) Neural correlates of auditory spatial processing in unilateral deafness: relation to sound localization in noise and cortical plasticity. Association for Research in Otolaryngology, Poster

Lee J, Han J-H, Lee H-J (2021) Development and Evaluation of stimuli for musical perceived emotions in people with hearing loss. Korean Society for Music Perception and Cognition, Oral

Lee J, Han J-H, Lee H-J (2021) The evaluation of newly developed-musical stimuli for the investigation of musical perceived emotions in people with hearing loss. Korean Society of Otorhinolaryngology-Head and Neck Surgery, Poster

Han J-H, **Lee J**, Lee H-J (2021) Attentional modulation of auditory-evoked cortical responses for sound localization in noise in unilateral deafness. Korean Society of Otorhinolaryngology-Head and Neck Surgery, Poster

Lee J, Han J-H, Lee H-J (2021) Cortical processing of frequency change reflected by event-related potentials is modified by long-term musical training. Korean Society of Otorhinolaryngology-Head and Neck Surgery, Poster

Lee J, Han J-H, Lee H-J (2020) Auditory cortical activity evoked by frequency change is altered by long-term musical training, Korean Society for EEG and Neurophysiology, Poster

Lee J, Han J-H, Lee H-J (2019) Noise effect on the auditory evoked responses to different

sound locations in unilateral deafness: Hemispheric asymmetry and relationship to speech perception. Society for Neuroscience, Poster

Lee J (2018) Timing-related neural activity in the dorsal hippocampus and posterior thalamus in rats. Korean Otological Society, Oral

Lee J, Chang SY (2017) Light stimulation can modulate harmaline-induced tremor in rats. Minnesota Neuromodulation Symposium, Poster

Lee J, Kim IY, Chang SY (2016) Harmaline-induced tremor in large animals. Society for Neuroscience, Poster

Lee J, Chang SY (2016) The activity of ventro-lateral thalamus in rats with harmaline-induced tremor. Department of Neurologic Surgery, Mayo clinic, Poster

Lee J, Jung MW (2012) Timing-related neural activity in the dorsal hippocampus and posterior thalamus in rats. Society for Neuroscience, Poster

Lee J, Kim J, Jung MW (2012) Discharge characteristics of neurons in posterior thalamus and hippocampus in rats during time interval discrimination task. The Korean Society for Brain and Neural Science, Poster

Lee J, Kim J, Jung MW (2012) Posterior thalamic and hippocampal neuronal activity during time interval discrimination task in rats. The 5th Brain Research Center-United Kingdom Joint Symposium on Neuroscience, Poster

Lee J, Jung MW (2012) Characteristics of posterior thalamic and hippocampal neuronal activity during time interval discrimination task in rats. Ajou Brain Conference, Oral presentation

Lee J, Jung MW (2010) Characteristics of thalamic and hippocampal neuronal activity in rats performing a temporal discrimination task. The Korean Society for Brain and Neural Science, Poster

Lee J, Jung MW (2010) Neural activity in the hippocampus and posterior thalamus during interval timing behavior in rats. Society for Neuroscience, Poster

Editorial

2024 – Present Consulting Editor, *Music Perception*

Peer Review / Reviewing Activity

International Journal of Developmental Neuroscience (2026, 1 manuscript)

Ear and Hearing (2025, 1 manuscript)

PLOS One (2025, 2 manuscripts)

Frontiers in psychology (2023, 1 manuscript)

Mentoring

2023 – Present Sungmin Jo, PhD student, Advisor

2021 – 2022 Eunhye Park, research assistant, Advisor

2015 – 2017 Students Receive Summer Fellowships to Mayo Clinic, Advisor